Tata Power EV Charging Network

Bengaluru - Utilisation, Customer Demand & Expansion ROI

Data Analyst Insights Report

114,000 Charging sessions	**2.86 GWh** Energy delivered	**Rs 4.97 Cr** Session revenue	**8,000** Customers
65 Stations / 18 districts	**390** Plugs	**98** Proposed projects	**41.4%** Avg 5-yr ROI

Data period: 11 Aug 2024 - 08 Nov 2024 (90 days). Tools: Excel (Power Query, Power Pivot data model, DAX measures, PivotTables, slicers, dashboards) and Python (pandas, matplotlib) for validation and this report.

1. Executive summary

Tata Power is planning the next phase of its EV charging network in Bengaluru. This analysis combines three months of operational data from 65 stations (all five operators in the city) with a financial model of 98 proposed sites to answer three questions: how the current network is used, who the customers are, and where new capital should go.

- **Demand is steady, and it exceeds capacity.** The network handles about **1,267 sessions a day**, flat from August to November. Every plug serves ~292 sessions, and utilisation is almost the same at every station (272-310 per plug). That points to a network that is capacity-bound rather than location-bound.
- **The city needs ~1,067 plugs but has 390.** The gap is **677 plugs** (the network would need to be 2.7x its current size). Five districts (Jayanagar, Rajajinagar, HSR Layout, Indiranagar, Koramangala) account for 52% of it.
- **High-income customers are worth 3x more.** They are 15% of sessions but 31% of revenue (Rs 12,979 per customer vs Rs 4,109 for Low-Mid).
- **AC chargers are where the returns are.** All 54 High-priority projects are AC Level-2. On average, AC projects return 86% over five years and DC projects -26%. 37 of the 98 projects lose money over five years, nearly all of them DC.
- **Recommended first wave:** the 54 High-priority projects. They cost **Rs 4,980 L** (Rs 49.8 Cr), generate Rs 1,907 L EBITDA a year, average a 93% 5-year ROI and pay back in about 2.6 years as a portfolio.

2. Data & method

Dataset	Rows	Grain / purpose
charging_sessions	114,000	One row per charging session (fact table)
charging_stations	65	Station master: district, operator, plugs, lat/long
customers	8,000	Customer income tier, car model, battery size
districts	18	Projected EVs, plugs and regular customers per district
roi_bengaluru_districts	98	CAPEX / revenue / payback / ROI per proposed project

Workflow. I loaded the CSVs through Power Query and related them in the Power Pivot data model (sessions -> stations -> districts; sessions -> customers; ROI -> districts). DAX measures (Total Sessions, Sessions per Plug, Revenue per Customer, Plug Gap, Avg Payback, Avg 5-yr ROI and others) feed 33 PivotTables and four dashboards with Month, Income Tier, Charger Type and Priority Tier slicers. I added a Battery Band column and Month/Year columns for segmentation. The figures were then re-computed independently in Python to check them.

Data quality checks

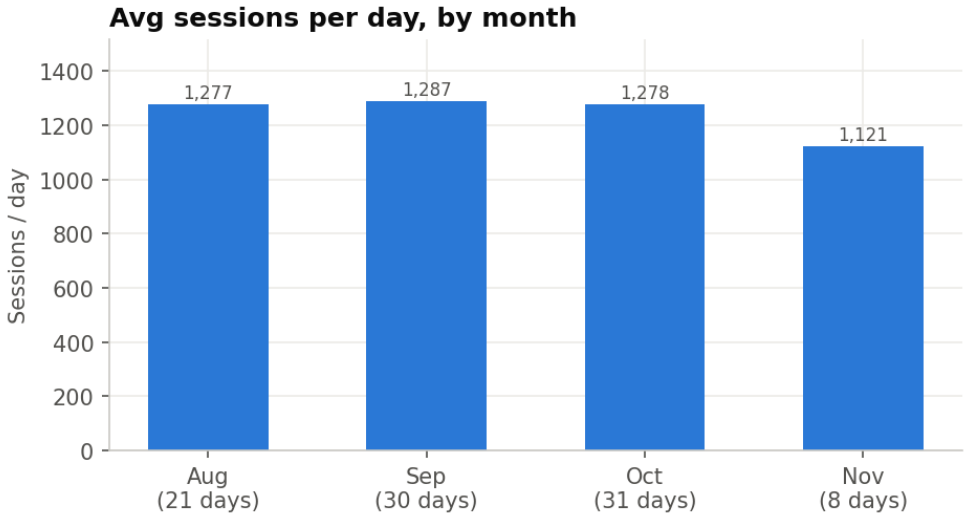
Check	Result
Sessions with unknown station	0
Sessions with unknown customer	0
Duplicate session_id	0
total_cost != kWh x tariff (>Rs 0.01)	0
Distinct plugs_count values	1

- **Reclassified ROI tiers:** 23 projects with a negative 5-year ROI had been tagged 'Medium'. I moved them to 'Low', and added a Capitalized_Land_Lease_10yr column so the CAPEX formula

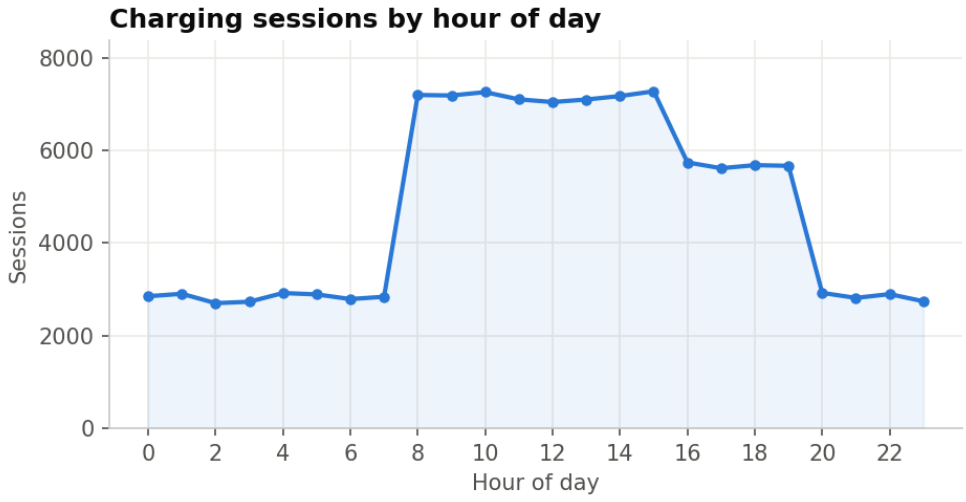
can be traced.

- **plugs_count is a placeholder:** every station has exactly 6 plugs. Plug-based metrics depend on this value, so it should be checked in the field.
- **Partial months:** the sessions cover 11 Aug to 08 Nov 2024. August and November are incomplete, so a raw monthly total makes November look like a collapse. Section 3 uses sessions per day instead.

3. Network utilisation

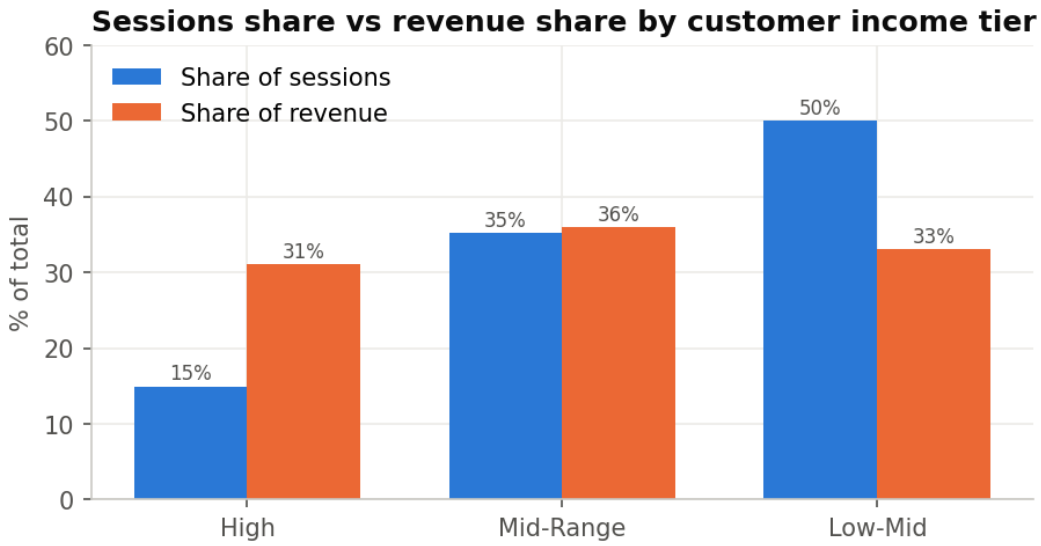


- Daily demand is flat at ~1,287 sessions a day in September and October. The low November total (8,970 sessions) comes from 8 days of data, not falling demand.
- The average session delivers 25.1 kWh and earns Rs 436. Per station, the network averages 1,754 sessions and 43,996 kWh.
- Utilisation is very even: 272-310 sessions per plug, a coefficient of variation of only 2.6%. No station sits idle. The best way to grow volume is to add plugs, not to rebalance existing load.



- Daytime peaks fall at 08:00, 10:00, 15:00 (~7,273 sessions in the busiest hour). The quietest hours are 02:00, 03:00, 23:00, at about 37% of peak.
- Off-peak (night) tariffs could move flexible fleet charging into the 23:00-05:00 trough and free daytime capacity.

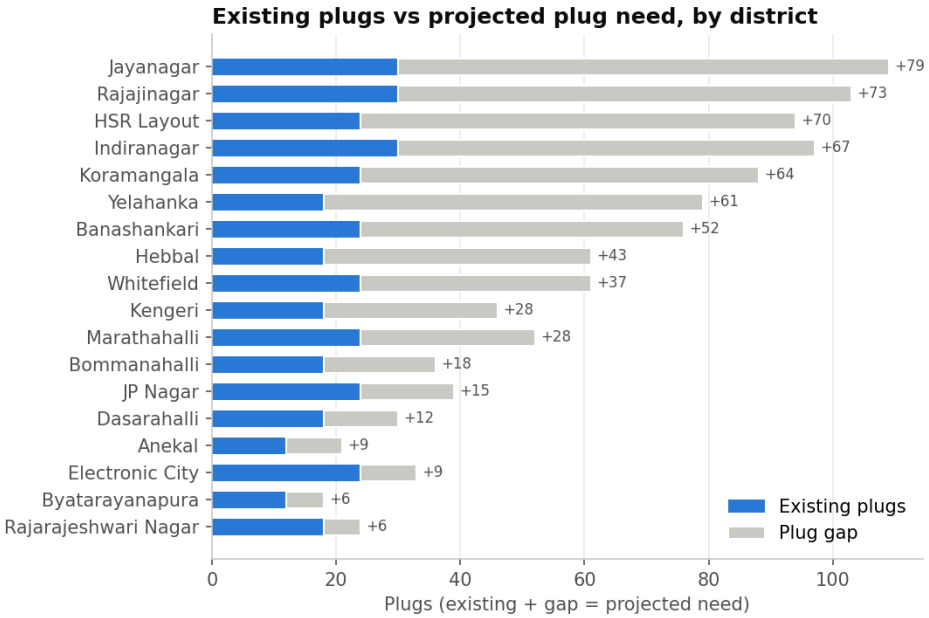
4. Customers & demand behaviour



Income tier	Customers	Sessions	Revenue	Avg kWh	Rev / customer
High	1,190	16,983	Rs 1.54 Cr	52.3	Rs 12,979
Mid-Range	2,820	40,059	Rs 1.79 Cr	25.6	Rs 6,331
Low-Mid	3,990	56,958	Rs 1.64 Cr	16.6	Rs 4,109

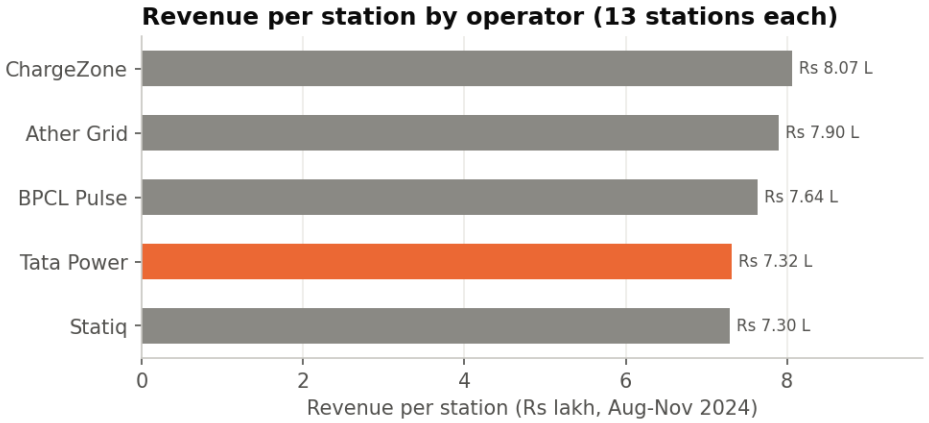
- Low-Mid customers are half of all sessions (50%) but bring in only 33% of revenue. Their vehicles have small batteries (2- and 3-wheelers, entry-level cars), so each session draws fewer kWh at lower tariffs.
- High-income customers charge 52.3 kWh per session, against 16.6 kWh for Low-Mid, and are worth 3.2x more each. Loyalty or subscription plans aimed at them would protect the most valuable revenue.
- Customers average 14.25 sessions each and Rs 6,212 in revenue. The most-used models are Ampere Nexus, Hero Electric Optima, Tata Tigor EV Base, all affordable, high-frequency vehicles.
- The mid tariff (Rs 18/kWh) carries 46% of sessions and 48% of revenue. It is the volume engine of the network.

5. Supply gap & competition



District	Tier	Stations	Proj. EVs	EVs / station	Plug gap
Jayanagar	Mid-Range	5	3,600	720	79
Rajajinagar	Mid-Range	5	3,400	680	73
HSR Layout	Mid-Range	4	3,100	775	70
Indiranagar	High	5	3,200	640	67
Koramangala	High	4	2,900	725	64

- Supply is not tracking demand. Jayanagar (3,600 projected EVs) has 5 stations, the same as Indiranagar, while Yelahanka has ~867 EVs per station, the most underserved ratio in the city.
- Electronic City, JP Nagar, Rajarajeshwari Nagar and Byatarayanapura are close to their projected need, so new sites there have low priority.

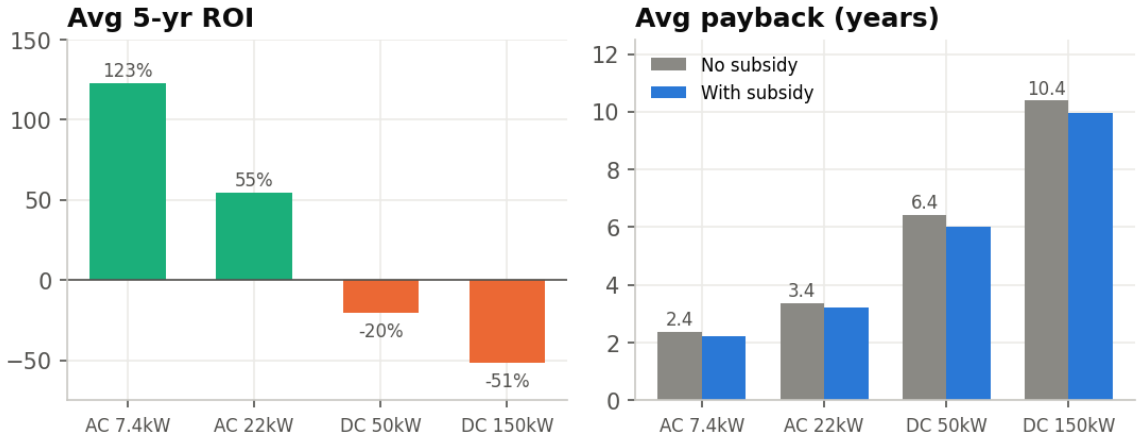


- The market is evenly split: each of the five operators runs 13 stations and takes 19-21% of revenue. Tata Power's share is 19.1%, Rs 7.32 L per station, compared with Rs 8.07 L for ChargeZone, the leader.
- Tata Power has no scale advantage yet. Winning the high-gap districts first is the clearest way to build share.

6. Expansion economics (98 proposed projects)

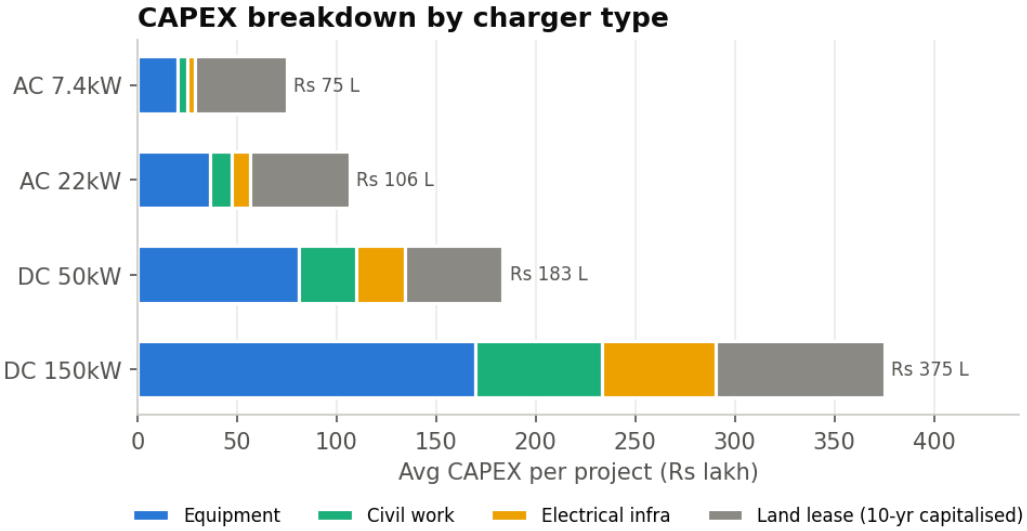
Rs 142 L	4.61 yrs	4.34 yrs	41.4%
Avg CAPEX / project	Avg payback	Avg payback w/ subsidy	Avg 5-yr ROI

Proposed projects: returns by charger type



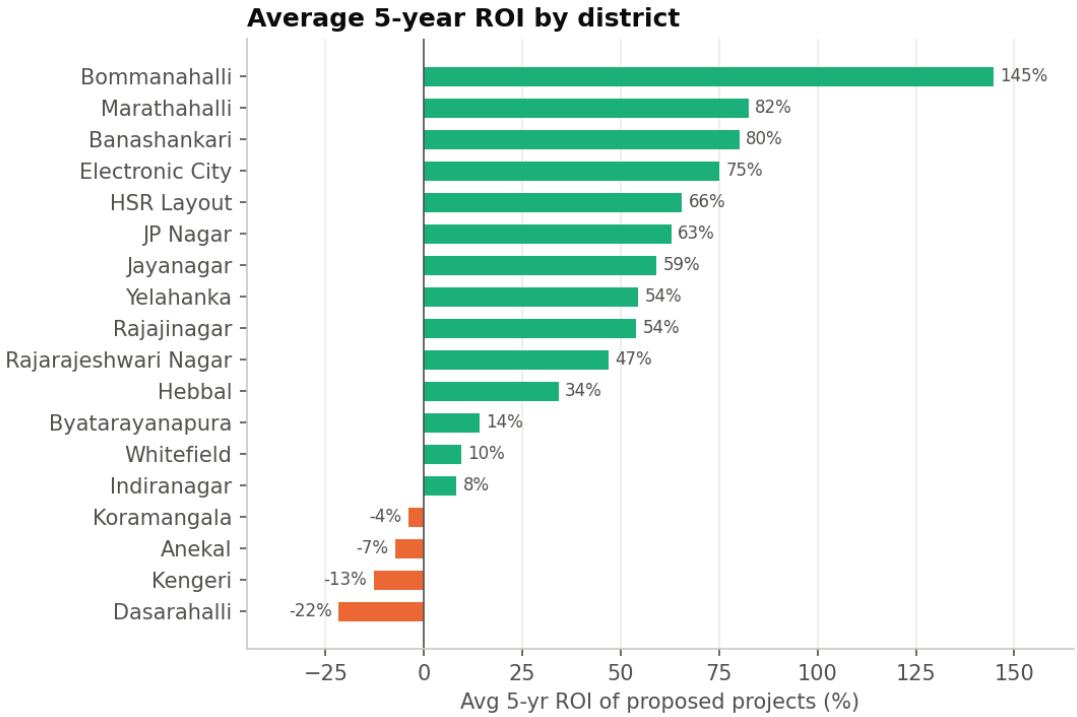
Charger type	Projects	Avg CAPEX	Payback	w/ subsidy	5-yr ROI	High tier
AC Level-2 (7.4 kW)	27	Rs 75 L	2.4 y	2.2 y	123%	27
AC Level-2 (22 kW)	32	Rs 106 L	3.4 y	3.2 y	55%	27
DC Fast (50 kW)	32	Rs 183 L	6.4 y	6.0 y	-20%	0
DC Ultra-Fast (150 kW)	7	Rs 375 L	10.4 y	10.0 y	-51%	0

- AC Level-2 (7.4 kW) is the clear winner: 123% average ROI and a 2.4-year payback. DC Ultra-Fast costs 5.0x as much and takes 10.4 years to pay back.
- Subsidies take ~0.5 years off payback for the 55 eligible projects (total subsidy Rs 781 L). Every eligible project should apply.



- Ten years of land lease, capitalised under Ind AS 116, is ~54% of an AC project's CAPEX. Choosing low-lease sites (residential complexes, bus depots) has a large effect on ROI.

7. Where to invest



- Mid-Range districts give the best returns (avg ROI 67%). High-income districts return only 4%, because their proposals lean towards expensive DC chargers (avg CAPEX Rs 230 L vs Rs 118 L) on high-lease land.
- By site type, Residential Complexes (61% ROI) and Bus Depots (52%) do best. Malls / Retail is the weakest (28%).

Top 10 projects by 5-year ROI

Project	District	Site	Charger	CAPEX	Payback	ROI
BLR-PRJ055	Marathahalli	Mall / Retail	AC (7.4 kW)	Rs 93 L	1.6 y	213%
BLR-PRJ086	Bommanahalli	Office Complex	AC (7.4 kW)	Rs 59 L	1.8 y	183%
BLR-PRJ041	HSR Layout	Public Parking	AC (7.4 kW)	Rs 85 L	1.8 y	183%
BLR-PRJ051	Banashankari	Office Complex	AC (7.4 kW)	Rs 102 L	1.8 y	183%
BLR-PRJ065	Yelahanka	Petrol Station	AC (7.4 kW)	Rs 60 L	1.8 y	178%
BLR-PRJ048	Banashankari	Public Parking	AC (7.4 kW)	Rs 85 L	1.9 y	163%
BLR-PRJ085	Bommanahalli	Petrol Station	AC (7.4 kW)	Rs 60 L	1.9 y	163%
BLR-PRJ064	Electronic City	Petrol Station	AC (7.4 kW)	Rs 82 L	1.9 y	161%
BLR-PRJ039	HSR Layout	Public Parking	AC (7.4 kW)	Rs 68 L	2.1 y	143%
BLR-PRJ036	Rajajinagar	Bus Depot	AC (7.4 kW)	Rs 83 L	2.1 y	139%

8. Recommendations

1. Fund the 54 High-priority AC projects first

Rs 4,980 L CAPEX, Rs 1,907 L EBITDA a year, about a 2.6-year portfolio payback. Phase them by plug gap, starting with Jayanagar, Rajajinagar, HSR Layout and Yelahanka.

2. Put DC fast charging on hold or redesign it

The 39 DC projects average a -26% ROI. Keep DC only for highway or corridor sites and fleet contracts with guaranteed volume, or renegotiate the lease and equipment terms.

3. Fix the high-income district plan

Indiranagar, Koramangala and Whitefield have strong demand but weak project ROI. Rework those proposals around AC chargers at residential and office sites instead of DC at malls and petrol stations.

4. Grow revenue per session

Offer loyalty or subscription plans for High and Mid-Range customers, use time-of-day pricing to fill the night trough, and push fleet (2- and 3-wheeler) partnerships for volume.

5. Claim every subsidy

55 projects are eligible (FAME-II or state EV policy). The subsidy cuts their payback by ~0.5 years.

6. Improve the data

Check plugs_count in the field, collect session duration and idle time to measure real utilisation, and extend the history to 12 months to capture seasonality.

Limitations

The session data is ~3 months long and includes all operators, not only Tata Power. ROI figures are planning-model projections, not actuals. EBITDA-based ROI leaves out tax, depreciation and the time value of money, so an NPV/IRR view is a natural next step.